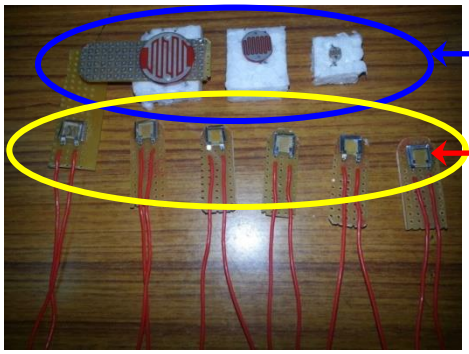
Photopatternable Silver and Photoconductor (CdS) thick film pastes for Photo Sensors

In India, many industries and primer institutions are working toward the new technique of selective deposition with improved performance and reduced cost in addition to minimizing the amount of material usage. Cosidering an imperitive need of current photoconductor market, the proposed project create great impact in the field of photosensor by selective patterning technology. The later is extremely important for the development of indigenous hardware capabilities for the fabrication of Sensors and the future sensor based system . Huge untapped market for photosensor in India is a well-known fact. An Indian endeavour for low cost sensor, viz., 'microphotosensor ', may also benefit from the miniaturization achieved by such circuits.

C-MET have developed an important step in the advancement of photoconductor technology offering the fabrication of simple process, potentially very low cost, miniature micro-photoconducting element, which can allow selective transfer and accurate registration. The prototypes photosensors have also been developed based on the photopatternable CdS thick film paste useful in the area of automatic car light control and digital object counters. In this context, an Indian endeavor of low cost photosensor, viz., 'microphotosensor', developed at C-MET may also benefit from the miniaturization achieved by such circuits. This photosensor can be called as micro-photosensor due to the micron dimensions of the line and space patterns.

Product specification



Applications

- *Automatic Headlight Dimmer*
- *Night Light Control*
- *Oil Burner Flame Out*
- *Street Light Control*
- *Absence / Presence (beam breaker)*
- *Position Sensor*

Specific use of the product/process in applications like Applicable

a) Auto Car Dimmer

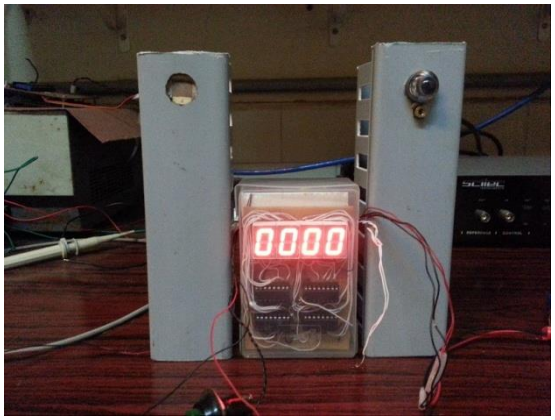
Properties	Values
Application	Auto Car Dimmer
Type	Resistive
Material	Activated CdS
Fabrication Process	Photopatterned electrode
Electrode spacing	100 micron
Total length	14mm
Device dimension	6 mm X6 mm
Spectral range	Visible (550nm peak)
Dark resistance	2.5Mohm ±10k
Light resistance	100kOhm±10k
Sensitivity	Three order
Response time	4.5ms
Power Supply	12 Volt (5 Amp) battery
Distance	25 meter
Light intensity (Standard 55 Watt bulb)	Upper light intensity (4500Lux) ±100 lux Lower light Intensity (44 lux) ± 8 lux

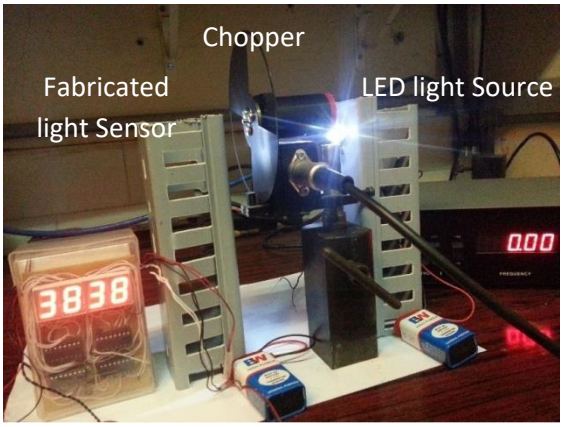
b)Counter Application

Properties	Values
Application	Counter
Type	Resistive
Material	Activated CdS
Fabrication Process	Photopatterned electrode
Electrode spacing	100 micron
Total length	140mm
Device dimension	12mm X14 mm
Spectral range	Visible (550nm peak)
Dark resistance	1.8Mohm ±10k
Light resistance	100kOhm±10k
Sensitivity	Three order
Response time	4.5ms
Light source	LED (1. 5 Volt x 2 AAA battery
Sensor and Circuit Power supply	9V volt battery
Distance	5 cm to 1000 cm
Speed	From 1Hz to 220 Hz
Long distance / remote application	Laser as a source (tested at distance100 meter)

Test performed at laboratory level

Currently, photodiode is used in such application due to speed limitation of LDR for such application (20 Hz).





High speed counter prototype: (1 Hz to 220 Hz)